



School of Computer & Systems Sciences

CS 760--NATURAL LANGUAGE PROCESSING

Monsoon -2025

Date: 12 December 2025

Time: 10:00 AM-1:00 PM

(50 marks)

NOTE: Questions 1 to 5 carry 2 marks each. All are mandatory **(2*5 =10 marks)**

Q1. The history of NLP generally starts in the _____

- a) 1960
- b) 1970
- c) 1950
- d) 1940

Q2. What is the primary purpose of Named Entity Recognition (NER) in NLP?

- a) To generate random text
- b) To improve text readability
- c) To recognize and classify named entities (e.g., names, locations)
- d) To perform tokenization

Q3. Epistemology is the _____

- a) An area of philosophy concerned with what there is in the universe.
- b) The philosophical study of being.
- c) The branch of philosophy that describes different entities and categories within reality.
- d) The branch of philosophy that examines the nature, origin, and limits of knowledge.

Q4. What is Morphological Segmentation?

- a) Does Discourse Analysis
- b) Separate words into individual morphemes and identify the class of the morphemes
- c) It is an extension of propositional logic
- d) None of the mentioned

Q5 Which NLP task aims to determine the correct meaning of a word based on its context?

- a) Sentiment Analysis
- b) Named Entity Recognition
- c) Word Sense Disambiguation
- d) Part-of-Speech Tagging

Note: Questions 6 to 11 carry 5 marks each. Attend any 4 questions (4 * 5 = 20 marks)

~~Q6.~~ Define Large Language Models (LLMs). Describe their training process, applications in NLP, and challenges associated with their use.

~~Q7.~~ Why is NLP called a direct applied area of AI? Justify your answer.

~~Q8.~~ What is Word Sense Disambiguation (WSD)? Why is it important in NLP?

Q9. What is sentiment analysis? Explain its machine learning-based approach and lexicon-based approach with suitable examples.

~~Q10.~~ Differentiate between a POS Tagger and a Parser with a suitable example.

~~Q11.~~ Explain the concept of Word to Vector (Word2Vec). How does it capture the semantic meaning of words?

Note: Questions 12 to 14 carry 10 marks each. Attend any 2 questions (2 * 10 = 20 marks)

Q12. How are Semantic Role Labeling (SRL) and Named Entity Recognition (NER) similar and different? Give a suitable example.

Q13. What are the key parameters of Artificial Intelligence (AI) with respect to Natural Language Processing (NLP)? Explain any four parameters with examples.

Q14. What is Text summarization? Describe the complete process of Automated Text Summarization in detail.